# P A R T I

## HISTORY AND ORGANISATION.

### History of the Pharmaceutical Industry.

The chemical-pharmaceutical industry has sprung up from more than one root. First of all, it had its origin in those old-famed places where remedies were first prepared, viz. the pharmacies. These participated to a great extent in the development of modern chemistry and they frequently proceeded to the production of specialties. But, while this development set in only at the beginning of the 19th century to a remarkable extent, efforts were already made during the later middle-ages to produce basic materials to be used in pharmacies, and perhaps also finished medicinal preparations.

The real history of development begins with the erection of small and smallest enterprises, which produced pharmaceuticals on the basis of scientific research work. We may begin with Johann Bartholomäus Trommsdoff (1770 - 1837) who erected the first factory of pharmaceutical and chemical products as early as 1813. He may be considered the representative of those contemporary pharmacists who were so much touched by the spirit of enlightenment of their age that they consciously put aside the nimbus and the mystery which characterized the pharmaceutical profession at that time, and devoted themselves passionately to natural science. Therefore they very often did not remain mere pharmacists but became teachers of natural science especially of chemistry, even if a special chair - as held by Trommsdorff - was not available to them. They published their discoveries in printed treatises and thus contributed at least indirectly to the creation of modern chemistry and thereby to the foundation of the respective industries. The particular importance of Trommsdorff is that in 1792 he erected an institute in Erfurt (probably the first in Germany), in which he trained young people in natural science and also in pharmacy.

In this connection, we may mention the work of Friedlieb Ferdinand Runge (1795 - 1867) who, when analysing coal-tar, discovered besides aniline and other components also phenol (1834) which he called carbolic acid. His activity is but an example of how strongly the discoveries of some individuals in natural sciences have influenced the rise of the chemical and also of the pharmaceutical industries. For just phenol serves not only as an antiseptic, it is at